Final Stages of the Plasma Column Evolution in the Plasma-Focus PF1000 Device

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***Abstract*—A frame high-speed camera was used to visualize the evolution of the final stages of the plasma-focus phenomenon. The frame images were taken with a 1-ns exposure time in visible continuum radiation with a narrow band interference filter. These features of the diagnostic equipment enabled the detailed study of the collapse phase as well as the plasma column creation and disruption processes with high temporal and spatial resolution.**

***Index Terms*—High-speed photography, MHD instabilities, plasma-focus.**

THE MEASUREMENTS were performed on the PF1000 plasma-focus device, installed at the Institute of Plasma Physics and Laser Microfusion, Warsaw, Poland. It is equipped with Mather-type electrodes. The outer grounded electrode of the 368-mm diameter is made of 24 stainless steel rods. The inner copper electrode (240-mm diameter and 450 mm in active length) is dynamically charged by a positive voltage. The cylindrical alumina insulator is fixed between the electrodes and is 113 mm in operating length. These parts are located inside the experimental chamber filled with deuterium of 1–5 torr. The condenser bank was charged up to 40 kV. The total energy stored in the bank was in the range of 900–1000 kJ, while the maximum of the discharge current was 3 MA.

The diagnostic system was based on the four-frame high-speed QUADRO camera equipped with a digital readout [1]. The sequences of plasma images were recorded with an exposure time of 1 ns and delay between subsequent frames in the range of 10–50 ns. An interference filter ($\lambda_{\mathrm{MAX}} = 593$ nm, $FWHM = 6$ nm, $T = 30\%$) was put into the optical path. Its spectral position enabled us to record the plasma images in region nearly with devoid of line radiation. Therefore, the QUADRO camera recorded only continuum radiation (almost pure bremsstrahlung), which allowed us to determine the rough relation between an intensity of the plasma radiation (J) and the electron density (n_e), $J \sim n_e^2$, and reconstruct spatial electron density distributions.

The equipment described above was used for investigation of the final stages evolution of the plasma-focus phenomenon. As it was shown in [2], their characteristic data (e.g., a radial velocity of a plasma sheath, the minimum radius of a plasma column and a dynamics of a plasma column disruption) determine level of the neutron yield. Hence, the precise qualitative and quantitative analysis of these processes is of interest.

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The occurrence of bad and good shots can be distinguished in Fig. 1(a). Bad shots are characterized by a lack of plasma sheath symmetry, the thicker plasma sheath and its lower dynamics.

The typical course of the plasma-focus phenomenon for shots with relative high neutron yield is shown in Fig. 1(b). The plasma column is created as a result of the collapse of the plasma sheath. In the case of the PF1000 device, the collapse phase lasts about 1.2 μs. The detailed investigation of the dynamics of the plasma sheath during this phase allowed us to determine certain regularities of this motion. First, the plasma sheath has the same axial velocity, v_z, during the whole collapse phase. This value depends on the particular working conditions and lies in the range of 0.7–0.9 $\times\ 10^7$ cm/s. Second, the initial value of the radial velocity, v_r, of the plasma sheath, taken from experimental data, is approximately equal to 0.5 v_z. This result is very close to the theoretical value calculated on the basis of shock wave diffraction on the inner electrode edge [3]. Third, the radial velocity of the plasma sheath in the vicinity of the inner electrode face is a linear function of the radius and its maximum value reaches 2×10^7 cm/s on the device axis. As a result of the radial compression of the plasma sheath, the plasma column is created on the device axis. The period of the plasma column creation is about 250 ns. After this time, the plasma column has about a 12 cm length. The long duration of the plasma column creation results in a disruption of the earliest created part of the plasma column before the creation process is finished. As a rule, the plasma column at any cross-section, after reaching the maximum compression state (the minimum radius, $r_{\min}$, of the plasma column in this state is about 0.7 cm), expands to a radius more than twice $r_{\min}$. This expansion lasts about 100 ns. Next, the development of the MHD $m = 0$ instability induces the gradual disruption of the plasma column.

Our investigation has shown, that two kinds of the plasma sheath disturbance can be distinguished [Fig. 1(c)], namely classical MHD $m = 0$ instability and singular large scale disturbance. Both kinds of disturbance occur in the same working condition of the PF-1000 device. The mechanism of the singular large scale disturbance is not clear and is to be investigated during future experiments. However, it should be emphasized, that for both types of disturbance the neutron yield is approximately the same.

Finally, it should be remarked that the investigation carried out by means of the high-speed frame photography revealed a number of the interesting insights about the final stages of the plasma-focus phenomenon. By comparing these results with ones obtained from smaller devices [2], [4], one can determine scaling laws for the increasing size of the plasma-focus devices.

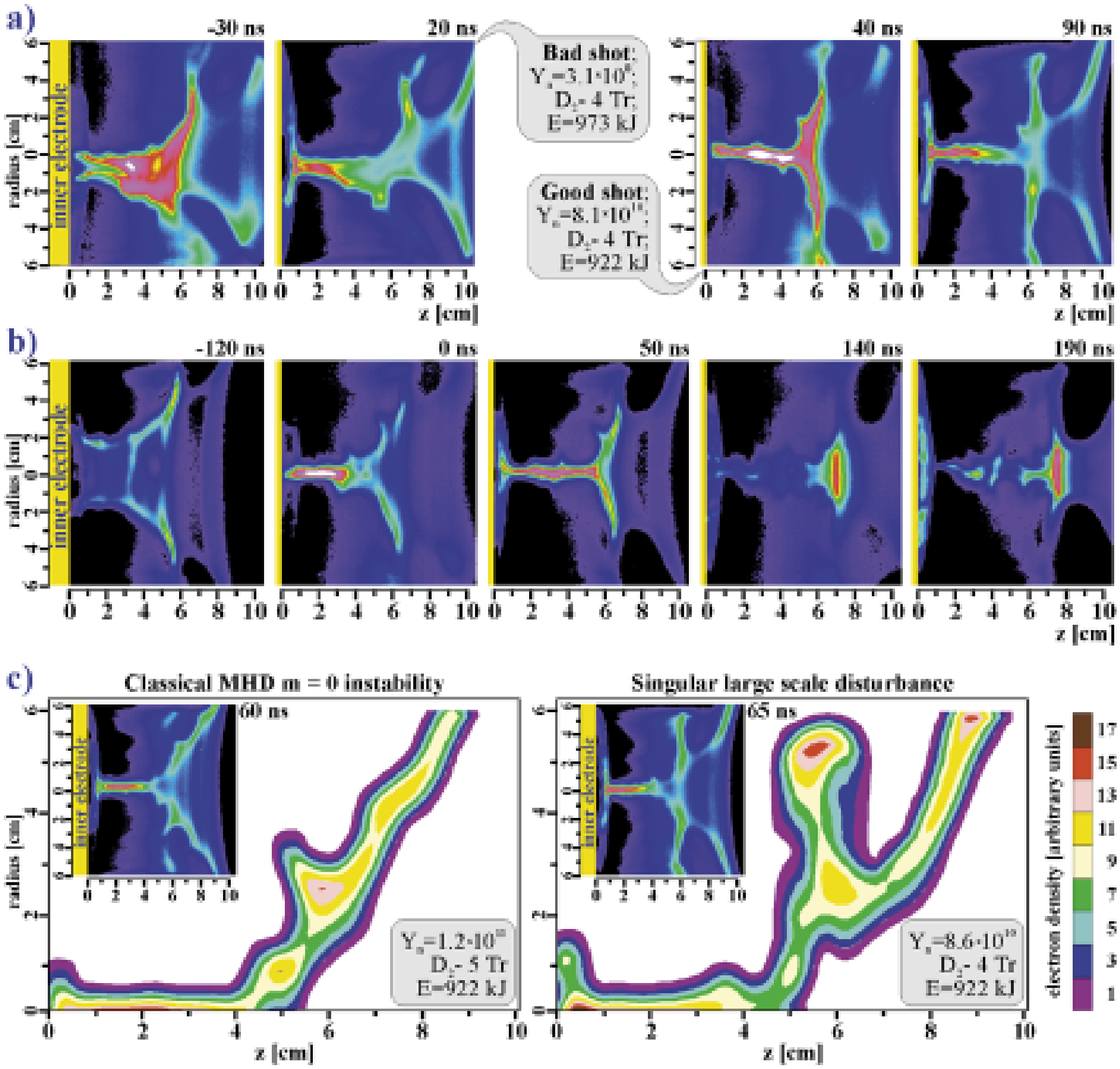

Fig. 1. Visualization of the final stages of the plasma-focus phenomenon. Multiframe images taken with 1-ns exposure time in visible continuum (593 nm) radiation, $t = 0$ refers to maximum compression. (a) Two-frames sequences obtained in shots with completely different neutron yield. (b) Typical course of the plasma column creation and disruption for $E = 930$ kJ, $p_{\mathrm{Dcut.}} = 2$, Tr and average $Y_n \sim 8 \times 10^{10}$. (c) Frame images and corresponding electron density distributions for two kinds of the plasma sheath disturbances recorded at shots with comparable neutron yield.

REFERENCES

[1] A. Krejči, K. Kolaček, J. Kravárik, P. Kubeš, M. Paduch, and K. Tomaszewski, "Evolution of gas-puff Z-pinch during implosion and pinch phase," *IEEE Trans. Plasma Sci.*, vol. 24, pp. 22–23, Feb. 1996.

[2] S. Czekaj, S. Denus, A. Kasperczuk, R. Miklaszewski, M. Paduch, S. Šledziński, K. Tomaszewski, Z. Wereszczyński, and J. Wolski, "Investigation of connections between the plasma column evolution and the neutron emission in the plasma-focus device," in *Proc. 11th Eur. Conf. on Controlled Fusion and Plasma Physics*, vol. 7D, part I, Aachen, Germany, 1983, pp. 469–472.

[3] G. B. Whitham, *Linear and Nonlinear Waves*. New York: Wiley, 1974.

[4] H. Herold, A. Jerzykiewicz, M. Sadowski, and H. Schmidt, "Comparative analysis of large plasma focus experiments performed at IPF Stuttgart and at IPJ Swierk," *Nucl. Fusion*, vol. 29, pp. 1255–1269, 1996.